Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	19 October 2026
Team ID	
Project Name	IMovies
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

Functional Requirements – Theatre Movies Ticket Booking App

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User & Admin Authentication	- User registration, customer login, and session persistence - Administrator authentication with role-based access control (RBAC) - Password encryption via bcryptjs and secure JWT token verification
FR-2	Movie & Theatre Catalog Discovery	- Publish and manage movie metadata (title, poster, duration, genres) (Admin) - Register cinema halls with tiered auditorium capacities (Admin)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		- Browse now-showing movies, search by title, and filter by genre (User) - Add and manage personal movie watchlists / favorites (User)
FR-3	Showtime Scheduling & Seat Management	- Schedule movie showtimes tied to specific dates, theatres, and pricing (Admin) - Automated generation of seating layouts across Lower, Middle, and Balcony tiers - Interactive seat grid with real-time atomic seat locking to prevent double-booking (User)
FR-4	Ticketing, Payments & Notifications	- Secure checkout supporting multiple payment methods (UPI, Card, Net Banking)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		- Generation of digital M-Tickets with unique references and scannable gate-entry QR tokens - Automated background email receipts, showtime reminders, and expired seat hold reclamation - Community movie reviews and star rating submission

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The platform should offer an intuitive, cinema-themed dark UI with clear interactive seating layouts, responsive navigation, and frictionless checkout across mobile and desktop devices.

NF R-2	Security	All data transfers must use HTTPS. Passwords must be hashed using bcryptjs, API endpoints protected by role-based access control (User and Admin tiers), and sessions validated via JWT tokens.
NF R-3	Reliability	Ensure atomic seat reservation, payment confirmation, and ticket issuance work reliably under peak traffic with a 0% double-booking collision guarantee.
NF R-4	Performance	Movie catalog and showtime pages should load within 2 seconds. Seat selection toggles and live availability updates should be near-instant (< 200ms).
NF R-5	Availability	The ticketing platform should maintain 99.9% uptime, with automated background services reliably reclaiming abandoned seat holds every 15 minutes.
NF R-6	Scalability	The system should support high-concurrency booking spikes during premiere release windows through a modular, domain-driven architecture and indexed MongoDB collections.